



Weather Company Data - Enhanced Forecast | Gridded & Polygonal Forecasts - v2

Domain Portfolio: Weather Imagery | Domain: Forecast | API Name: Gridded & Polygonal Forecasts - v2

Standard HTTP Cache-Control headers are used to define caching length. The TTL value is provided in the HTTP Header as an absolute time value using the “Expires” parameter.
Example: “Expires: Fri, 12 Jul 2013 12:00:00 GMT”.

Geography: Global

Attribution Required: NO

Attribution Requirements: N/A

Overview

The ‘Tiler’, and ‘Featurizer’ products support data visualization and analytics.

- Tiler provides gridded raster data, typically in tiles of 256x256 pixels at various levels of detail; a client-side SDK can use this data to create weather image tiles
- Featurizer provides geometric vector data, either a line or a polygon, indicating where meteorological values cross a particular threshold; this data can facilitate statistical analysis
- For additional details about Tiler and Featurizer please see the Weather Company Data | Common Usage Guide - Data Visualization - Weather Imagery:
 - <https://ibm.co/DVWCUG>

Using the Tiler and Featurizer products require a multi-step workflow to retrieve the necessary data for the specific product data request. Steps 2a, and 2b are dependent on which type you are requesting (i.e. Tiler Data or Featurizer Tile. Both step 2a, and step 2b require the ‘t’ parameter values as input into the ‘t’ parameter for the subsequent request (v2/tiler/data, and v2/featurizer/tile).

- **Step 1:** Get Tiler Info - Provides current dimensions ‘t’ and ‘rt’ parameter values on one or more products.
- **Step 2a:** Get Tiler Data - Provides tiles of meteorological data from one or more products.
- **Step 2b:** Get Featurizer Tile - Provides a polygon or line indicating when a product’s data has crossed a given threshold, in web-mercator projection.

URL Construction

Step 1: Get Tiler Info
Tiler Info: Required Parameters: products, apiKey=yourApiKey Optional Parameters: meta=true https://api.weather.com/v2/tiler/info?products=<productNumber>:<variableID>&apiKey=yourApiKey
The [v2/tiler/info?] request response provides the ‘t’ parameter value required as input for the subsequent [v2/tiler/data?] request as well as a subsequent [v2/featurizer/tile?] request. If the product is an ‘Observation/Current Condition’ type then it will return a ‘t’ parameter value; if the product is a ‘Forecast’ type then it will return both a ‘t’ parameter value and ‘rt’ parameter value. • Note: some exceptions may apply to the use of the ‘t’ parameter value and ‘rt’ parameter values; please see product specific details for all product specific required and optional parameters.
https://api.weather.com/v2/tiler/info?products=49:Temperatureheightaboveground&meta=true&apiKey=yourApiKey
Step 2a: Get Tiler Data
Tiler Data - Forecast: Required Parameters: products, rt, t, lod, x, y, apiKey=yourApiKey https://api.weather.com/v2/tiler/data?products=<productNumber>:<variableID>&rt=<rt>&t=<t>&lod=<lod>&x=<x>&y=<y>&apiKey=yourApiKey
https://api.weather.com/v2/tiler/data?products=49:Temperatureheightaboveground&rt=1474375500000&t=1474400700000&lod=2&x=0&y=0&apiKey=yourApiKey
Step 2b: Get Featurizer Tile

Featurizer Tile - Forecast: Required Parameters: product , rt , t , lod , x , y , apiKey=yourApiKey Optional Parameters: threshold <a href="https://api.weather.com/v2/featurizer/tile?product=<productNumber>:<variableID>&rt=<rt>&t=<t>&lod=<lod>&x=<x>&y=<y>&apiKey=yourApiKey">https://api.weather.com/v2/featurizer/tile?product=<productNumber>:<variableID>&rt=<rt>&t=<t>&lod=<lod>&x=<x>&y=<y>&apiKey=yourApiKey
https://api.weather.com/v2/featurizer/tile?product=49:Temperatureheightaboveground&rt=1474375500000&t=1474400700000&lod=2&x=0&y=0&threshold=280&apiKey=yourApiKey
Featurizer Feature (Native Resolution) - Forecast: Required Parameters: product , rt , t , apiKey=yourApiKey Optional Parameters: threshold <a href="https://api.weather.com/v2/featurizer/feature?product=<productNumber>:<variableID>&rt=<rt>&t=<t>&apiKey=yourApiKey">https://api.weather.com/v2/featurizer/feature?product=<productNumber>:<variableID>&rt=<rt>&t=<t>&apiKey=yourApiKey
https://api.weather.com/v2/featurizer/feature?product=49:Temperatureheightaboveground&rt=1474375500000&t=1474400700000&threshold=280&apiKey=yourApiKey

Product Elements & Definitions

Product Name	Product Number	Variable ID	Reasonable Threshold
gFOD Thunderstorm Potential	44	VAR0191209FROM25501surface12MinuteMaximum	2
Gridded Forecast on Demand: Index of thunderstorm potential, where 0 is No Thunder, 1 is Thunder Possible, 2 is Thunder Expected, 3 is Severe Thunderstorms Possible, 4 is Severe Thunderstorms Likely, 5 is High Risk of Severe Thunderstorms: forecasted for 12-hour windows, across a span of 15 days: 4-km resolution, with a new set of forecasts generated every hour.			
gFOD 24-Hour Maximum Temperature Forecast	45	VAR00222FROM25501heightaboveground24MinuteMaximum	280
Gridded Forecast on Demand: Maximum temperature in kelvins (K), forecasted for 24-hour windows, across a span of 15 days: 4-km resolution, with a new set of forecasts generated every hour.			
gFOD 24-Hour Minimum Temperature Forecast	46	VAR00223FROM25501heightaboveground24MinuteMinimum	280
Gridded Forecast on Demand: Minimum temperature in kelvins (K), forecasted for 24-hour windows, across a span of 15 days: 4-km resolution, with a new set of forecasts generated every hour.			
gFOD Cumulative 12-Hour Snowfall Forecast	47	VAR01224FROM25501surface12MinuteAccumulation	0.6
Gridded Forecast on Demand: Cumulative snowfall in meters (m), totaled from the hourly process time to the end of each forecast window; forecasted for 12-hour windows, across a span of 15 days: 4-km resolution, with a new set of forecasts generated every hour.			
gFOD 15-Minute Dewpoint Temperature Forecast	48	Dewpointtemperatureheightaboveground	280
Gridded Forecast on Demand: Dewpoint temperature in kelvins (K), forecasted for 15-minute windows, across a span of 7 hours: 4-km resolution, with a new set of forecasts generated every 15 minutes.			
gFOD 15-Minute Temperature Forecast	49	Temperatureheightaboveground	280

Gridded Forecast on Demand: Temperature in kelvins (K), forecasted for 15-minute windows, across a span of 7 hours: 4-km resolution, with a new set of forecasts generated every 15 minutes.			
gFOD 12-Hour Precipitation Forecast	50	VAR01225FROM25501surface12MinuteAccumulation	600
Gridded Forecast on Demand: Precipitation in millimeters (mm), forecasted for 12-hour windows, across a span of 15 days: 4-km resolution, with a new set of forecasts generated every hour.			
gFOD 15-Minute UV Index Forecast	51	VAR0191200FROM25501heightaboveground	5
Gridded Forecast on Demand: UV value, measuring ultraviolet (UV) exposure, where 0-2 is Low, 3-5 is Moderate, 6-7 is High, 8-10 is Very High, and 11 is Extreme; forecasted for 15-minute windows, across a span of 7 hours: 4-km resolution, with a new set of forecasts generated every 15 minutes.			
gFOD 15-Minute Radar Reflectivity Forecast	52	Basereflectivitysurface	25
Gridded Forecast on Demand: Precipitation forecast expressed as radar reflectivity (dbZ), commonly referenced as "future radar;" forecasted for 15-minute windows, across a span of 7 hours: 4-km resolution, with a new set of forecasts generated every 15 minutes.			
gFOD 1-Hour Sustained Wind Speed Forecast	53	Windspeedheightaboveground	9
Gridded Forecast on Demand: Sustained wind speed in meters per second (m/s), forecasted for 1-hour windows, across a span of 15 days: 4-km resolution, with a new set of forecasts every hour.			
gFOD 12-Hour Wind Chill Forecast	54	Windchillfactorheightaboveground12MinuteMinimum	260
Gridded Forecast on Demand: Wind chill in kelvins (K), forecasted for 12-hour windows, across a span of 15 days: 4-km resolution, with a new set of forecasts generated every hour.			
gFOD 12-Hour Heat Index Forecast	55	Heatindexheightaboveground12MinuteMaximum	300
Gridded Forecast on Demand: Heat index in kelvins (K), forecasted for 12-hour windows, across a span of 15 days: 4-km resolution, with a new set of forecasts generated every hour.			
gFOD Cumulative 12-Hour Precipitation Forecast	57	VAR01223FROM25501surface12MinuteAccumulation	600
Gridded Forecast on Demand: Cumulative precipitation in millimeters (mm), totaled from the hourly process time to the end of each forecast window; forecasted for 12-hour windows, across a span of 15 days: 4-km resolution, with a new set of forecasts generated every hour.			
gFOD 12-Hour Snowfall Forecast	58	VAR01227FROM25501surface12MinuteAccumulation	0.6
Gridded Forecast on Demand: Snowfall in meters (m), forecasted for 12-hour windows, across a span of 15 days: 4-km resolution, with a new set of forecasts generated every hour.			

gFOD 15-Minute Sustained Wind Speed Forecast	59	Windspeedheightaboveground	5
Gridded Forecast on Demand: Sustained wind speed in meters per second (m/s), forecasted for 15-minute windows, across a span of 7 hours: 4-km resolution, with a new set of forecasts generated every 15 minutes.			
gFOD 15-Minute Feels-Like Temperature Forecast	60	VAR00198FROM25501heightaboveground	280
Gridded Forecast on Demand: TWC-created "feels-like" temperature in kelvins (K), based on temperature, heat index, and wind chill, forecasted for 15-minute windows, across a span of 7 hours: 4-km resolution, with a new set of forecasts generated every 15 minutes.			
gFOD 1-Hour Temperature Forecast	61	Temperatureheightaboveground	280
Gridded Forecast on Demand: Temperature in kelvins (K), forecasted for 1-hour windows, across a span of 15 days: 4-km resolution, with a new set of forecasts generated every hour.			
gFOD 12-Hour UV Index Forecast	62	VAR0191200FROM25501heightaboveground12MinuteAverage	5
Gridded Forecast on Demand: UV index measuring ultraviolet (UV) exposure, where 0-2 is Low, 3-5 is Moderate, 6-7 is High, 8-10 is Very High, and 11 is Extreme; forecasted for 12-hour windows, across a span of 15 days: 4-km resolution, with a new set of forecasts generated every hour.			
gFOD 15-Minute Precipitation Type Forecast	63	Precipitationtypesurface	40
Gridded Forecast on Demand: Precipitation type, which is most often used in combination with base reflectivity; forecasted for 15-minute windows, across a span of 7 hours: 4-km resolution, with a new set of forecasts generated every 15 minutes.			
gFOD 12-Hour Sustained Wind Speed Forecast	64	Windspeedheightaboveground12MinuteAverage	9
Gridded Forecast on Demand: Sustained wind speed in meters per second (m/s), forecasted for 12-hour windows, across a span of 15 days: 4-km resolution, with a new set of forecasts every hour.			
gFOD 15-Minute Cloud Coverage Forecast	65	Totalcloudcoversurface	9
Gridded Forecast on Demand: Cloud coverage expressed as a percentage, 0 to 100, forecasted for 15-minute windows, across a span of 7 hours: 4-km resolution, with a new set of forecasts generated every 15 minutes.			
gFOD Cumulative 1-Hour Precipitation Forecast	67	VAR01223FROM25501surface0MinuteAccumulation	150
Gridded Forecast on Demand: Cumulative quantitative precipitation forecast (QPF) in millimeters (mm), totaled from the hourly process time to the end of each forecast window; forecasted for 1-hour windows, across a span of 15 days: 4-km resolution, with a new set of forecasts generated every hour.			

gFOD Cumulative 1-Hour Snowfall Forecast	68	VAR01224FROM25501surface0MinuteAccumulation	0.15
Gridded Forecast on Demand: Cumulative quantitative precipitation forecast (QPF) for snowfall in meters (m), totaled from the hourly process time to the end of each forecast window; forecasted for 1-hour windows, across a span of 15 days: 4-km resolution, with a new set of forecasts generated every hour.			
gFOD 12-Hour Tree Pollen Index Forecast	70	VAR0191216FROM25501surface12MinuteAverage	2
Gridded Forecast on Demand: Tree pollen index, where 0 is None / Out of Season, forecasted for 12-hour windows across a span of 15 days: 4-km resolution, with a new set of forecasts generated every minute.			
gFOD 12-Hour Grass Pollen Index Forecast	71	VAR0191217FROM25501surface12MinuteAverage	3
Gridded Forecast on Demand: Grass pollen index, where 0 is None / Out of Season, forecasted for 12-hour windows across a span of 15 days: 4-km resolution, with a new set of forecasts generated every minute.			
gFOD 12-Hour Aches and Pains Index Forecast	72	VAR0191219FROM25501surface12MinuteAverage	8
Gridded Forecast on Demand: Aches and pains index, indicating the weather's influence on an individual's potential to feel aches and pains, forecasted for 12-hour windows across a span of 15 days: 4-km resolution, with a new set of forecasts generated every 24 hours.			
gFOD 12-Hour Breathing Index Forecast	73	VAR0191220FROM25501surface12MinuteAverage	6
Gridded Forecast on Demand: Breathing index, indicating the weather's influence on a respiratory-sensitive individual's ability to breathe, forecasted for 12-hour windows across a span of 15 days: 4-km resolution, with a new set of forecasts generated every minute.			
gFOD 12-Hour Ragweed Pollen Index Forecast	74	VAR0191218FROM25501surface12MinuteAverage	4
Gridded Forecast on Demand: Ragweed pollen index, forecasted for 12-hour windows across a span of 15 days: 4-km resolution, with a new set of forecasts generated every minute.			
gFOD 12-Hour Golf Index Forecast	76	VAR0191225FROM25501surface12MinuteAverage	6
Gridded Forecast on Demand: Golf index, forecasted for 12-hour windows across a span of 15 days: 4-km resolution, with a new set of forecasts generated every hour.			

RMB 15-Minute Radar Reflectivity Forecast	84	reflectivity	dBZ	25
Radar Model Blend (RMB): Precipitation forecast expressed as radar reflectivity (dBZ), commonly referenced as "future radar;" forecasted at 15-minute intervals, across a span of 7 hours: 4-km resolution, with a new set of forecasts generated every 5 minutes. This product is an update to product 52 and should be used in its place and used in conjunction with Precipitation Type found in product 63v2				

Document Revision History

Revision	Date	Notes
1.1	March 6, 2025	Added new product Id 84; addition of Document Revision History